



NETWORK ANALYSIS USING WIRESHARK

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Introduction to Computer Networks

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APRIL 15, 2026

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1. Explain Why Wireshark is seeing Traffic when Network Applications not actively in use.

For an understanding of the traffic, it is imperative to know the address of the subject device; *in hoc casu*, 192.168.100.6.

At a first glance, the majority of the initial traffic seems to take place squarely inside the home network itself—deduced from the subnet mask of the subject device being 255.255.255.0—some involving the subject device, others not.

Further investigation of the captured packets led us to these conclusions:

- + The bulk of the traffic being between my mobile phone and my laptop device as I have them synchronized.
 - From this we can extrapolate that most traffic is the subject device reaching out to online servers to synchronize itself.
- + Other captured packets were broadcast or multicast transmissions from other [known and unknown] devices in the home network to the broadcast or multicast address, or from the subject device to the broadcast or multicast address.
- + As theorised from the first point, further down we come across transmission captured between the subjected device and the Microsoft Corporation; the subject device running the Windows OS for its operation.
- + Again, similar as the previous point, another captured transmission is between the subject device and Mauritius Telecom Ltd, the internet provider of the home network.
- + Lastly, the set of addresses whose provenance and purpose couldn't be easily divined have been resolved to leading to various CDNs around the world.

Knowing where the traffic goes to, we can now theorise as to their purposes; some involving synchronisation with servers and devices, others being responses to network requests from other devices, or background processes gleaning information from remote servers as pertaining to their purposes (looking for [automatic] updates and what not).

2. Choose 5 Protocols in the Traffic: What do each do? Why are each important to the Network?

The following 5 protocols have been identified from the captured network traffic.

- I. **SSDP (Simple Service Discovery Protocol)** is used to discover devices and services that are available on a local network. It operates on the basis of UPnP (Universal Plug and Play), which allows devices such as printers and smart TVs to find and communicate with each other when plugged in or turned on. In this capture (screenshot below), this can be observed in packet 4, where the device 192.168.8.131 sends an “M-SEARCH * HTTP/1.1” request to a multicast IP 239.255.255.250, asking what devices are on the network. In reply, in packet 9, the device 192.168.8.1 responds with “HTTP/1.1 200 OK,” to confirm its presence. SSDP is therefore important as it allows devices to automatically and quickly find each other without any user configuration. For instance, when connecting a phone to a Smart TV, SSDP is working to locate that TV on the network.

4	0.005279	192.168.8.131	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
5	0.008798	167.105.216.66	192.168.8.131	TLSv1.3	1441 Application Data
6	0.010323	192.168.8.131	167.105.216.66	TLSv1.3	128 Application Data
7	0.010566	192.168.8.131	167.105.216.66	TLSv1.3	146 Application Data
8	0.010806	192.168.8.131	167.105.216.66	TLSv1.3	644 Application Data
9	0.037386	192.168.8.1	192.168.8.131	SSDP	417 HTTP/1.1 200 OK

- II. **DNS (Domain Name System)** is used to translate domain names (e.g. google.com) into numerical IP addresses. Each device connected to the internet has a unique IP address which can be used by other machines to find that device (Quiroz-Vázquez, 2024). In the screenshot below, a DNS packet shows a device sending a query to find the IP address for “login.microsoftonline.com”. Shortly after, packet 16 shows the DNS server replying with a Standard query response providing the correct IP address. DNS is important as it removes the need for users to remember the numeric IP addresses, hence making it easier to use the internet. The example also shows how DNS works instantly enabling faster connection to online services.

15	0.346168	192.168.8.131	192.168.8.1	DNS	85 Standard query 0xa92b A login.microsoftonline.com
16	0.357290	192.168.8.1	192.168.8.131	DNS	327 Standard query response 0xa92b A login.microsoftonline.com CNAME lo

- III. **TCP (Transmission Control Protocol)** ensures that data is delivered reliably between devices on a network. It breaks data into packets, sends them across the network and then rearrange them in order while also checking for errors (e.g. if any packets are missing) before delivering them to the application (Yasar & Goss, 2025). In the screenshot below, TCP is shown in packets 2 and 3, where my laptop (192.168.8.131) and the server (167.105.216.66) exchange ACK (acknowledgement) packets to confirm data is received. TCP is important to enable secure web browsing, email and file transfers. Without TCP, applications would have to deal with missing packets, duplicates or disordered packets themselves.

1	0.000000	167.105.216.66	192.168.8.131	TLSv1.3	1454 Server Hello
2	0.000000	167.105.216.66	192.168.8.131	TCP	1454 443 → 60974 [ACK] Seq=1401 Ack=1 Win=16381 Len=1400 [TCP PDU reassembled in 5]
3	0.000060	192.168.8.131	167.105.216.66	TCP	54 60974 → 443 [ACK] Seq=1 Ack=2801 Win=255 Len=0
4	0.005279	192.168.8.131	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1

- IV. **TLSv1.3 (Transport Layer Security)** is a cryptographic protocol that encrypts data during transmission, hence enabling secure transmission between devices (Cloudflare, nd). In this capture, TLSv1.3 is seen in packets 5 and 6, for example, where my laptop (192.168.8.131) communicates securely with a remote server (167.105.216.66). The process starts with a TLS handshake as seen in packet 1, where the client (my laptop) and server authenticate each other. TLS is essential to protect sensitive information such as passwords, emails and online

transactions, particularly involving personal financial details. The data is encrypted such that it cannot be read or tampered by unauthorised parties.

1	0.000000	167.105.216.66	192.168.8.131	TLSv1.3	1454 Server Hello
2	0.000000	167.105.216.66	192.168.8.131	TCP	1454 443 → 60974 [ACK] Seq=1401 Ack=1 Win=16381 Len=1400 [TCP PDU reassembled]
3	0.000060	192.168.8.131	167.105.216.66	TCP	54 60974 → 443 [ACK] Seq=1 Ack=2801 Win=255 Len=0
4	0.005279	192.168.8.131	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
5	0.008798	167.105.216.66	192.168.8.131	TLSv1.3	1441 Application Data
6	0.010323	192.168.8.131	167.105.216.66	TLSv1.3	128 Application Data

- V. **UDP (User Datagram Protocol)** is used for the same purpose as TCP in sending data from one device to another in a network. However, unlike TCP, UDP is a fast transport layer protocol favouring speed over reliability. It sends data immediately without setting up a prior connection. It also does not check for errors or ensure packets are in order, which makes it less reliable than TCP but faster (Yasar & Goss, 2025). Therefore, UDP is usually used for streaming, gaming and voice calls, where speed is more important than accuracy. The screenshot shows that UDP packets are sent between 192.168.8.131 and external IPs on port 443 with small sizes, showing fast data transfer without a connection. UDP is important to networks as it reduced delays and thus allows quicker communication between devices.

742	24.993574	40.104.14.194	192.168.8.131	UDP	84 443 → 62341 Len=42
696	24.914217	192.168.8.131	23.15.139.138	UDP	92 49438 → 443 Len=50
694	24.913871	192.168.8.131	40.104.14.194	UDP	98 62341 → 443 Len=56

3. How to tell if a Packet is sent by Your Computer?

Are there any that are neither sent by nor destined specifically for Your Computer? Why are these?

How can you tell if a packet is sent by your computer?

Use of **ipconfig** command in the command prompt of windows to know what the IP address of my device is:

IPv4 Address. : 192.168.100.37

In the packet capture file, each packet with the source column as 192.168.100.37 is considered to be a packet that is being sent by my computer because this is my IP address and it is the source address in these cases.

No.	Time	Source	Destination	Protocol	Length	Info
52	11.327067	172.217.170.206	192.168.100.37	TCP	66	443 → 18368 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
53	11.327178	192.168.100.37	172.217.170.206	TCP	54	18368 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
54	11.327495	142.251.156.119	192.168.100.37	QUIC	82	Initial, SCID=f1c7e881ae6280c8, PKN: 1, ACK
55	11.327760	192.168.100.37	172.217.170.206	TLSv1.3	1875	Client Hello (SNI=www.youtube.com)
56	11.328571	142.251.156.119	192.168.100.37	QUIC	1292	Initial, SCID=f1c7e881ae6280c8, PKN: 2, ACK, PADDING
57	11.328800	142.251.156.119	192.168.100.37	QUIC	1292	Initial, SCID=f1c7e881ae6280c8, PKN: 3, CRYPTO, PADDING
58	11.329123	142.251.156.119	192.168.100.37	QUIC	1292	Initial, SCID=f1c7e881ae6280c8, PKN: 4, CRYPTO, PADDING
59	11.332565	192.168.100.37	142.251.156.119	QUIC	1292	Initial, DCID=f1c7e881ae6280c8, PKN: 3, ACK, PADDING
60	11.341598	142.251.156.119	192.168.100.37	TCP	66	443 → 47098 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
61	11.341733	192.168.100.37	142.251.156.119	TCP	54	47098 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
62	11.342112	216.58.223.106	192.168.100.37	TCP	66	443 → 33401 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
63	11.342198	192.168.100.37	216.58.223.106	TCP	54	33401 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
64	11.342216	192.168.100.37	142.251.156.119	TLSv1.3	1778	Client Hello (SNI=www.google.com)
65	11.342649	192.168.100.37	216.58.223.106	TLSv1.3	1894	Client Hello (SNI=oathaccountmanager.googleapis.com)
66	11.352877	142.251.216.3	192.168.100.37	TCP	70	443 → 20725 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
67	11.353016	192.168.100.37	142.251.216.3	TCP	54	20725 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0

Applying a display such as **ip.src == 192.168.100.37** will only show all packets that is being sent by my computer. This filter can be useful in analyzing packets for any required conditions.

ip.src == 192.168.100.37						
No.	Time	Source	Destination	Protocol	Length	Info
68	11.353391	192.168.100.37	142.251.216.3	TLSv1.3	1889	Client Hello (SNI=clientservices.googleapis.com)
71	11.371215	192.168.100.37	142.251.156.119	QUIC	79	Handshake, DCID=f1c7e881ae6280c8
80	11.443110	192.168.100.37	142.251.156.119	QUIC	82	Handshake, DCID=f1c7e881ae6280c8
83	11.443956	192.168.100.37	142.251.156.119	QUIC	81	Handshake, DCID=f1c7e881ae6280c8
84	11.445300	192.168.100.37	142.251.156.119	QUIC	125	Handshake, DCID=f1c7e881ae6280c8
85	11.445519	192.168.100.37	142.251.156.119	QUIC	115	Protected Payload (KP0), DCID=f1c7e881ae6280c8
86	11.445693	192.168.100.37	142.251.156.119	QUIC	71	Protected Payload (KP0), DCID=f1c7e881ae6280c8
87	11.446134	192.168.100.37	142.251.156.119	QUIC	1288	Protected Payload (KP0), DCID=f1c7e881ae6280c8
88	11.446188	192.168.100.37	142.251.156.119	QUIC	1292	Protected Payload (KP0), DCID=f1c7e881ae6280c8
89	11.446202	192.168.100.37	142.251.156.119	QUIC	106	Protected Payload (KP0), DCID=f1c7e881ae6280c8
91	11.463308	192.168.100.37	142.250.110.84	TCP	54	53421 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
93	11.463479	192.168.100.37	142.250.110.84	TCP	54	30236 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
95	11.463587	192.168.100.37	142.250.110.84	TCP	54	20909 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
96	11.463804	192.168.100.37	142.250.110.84	TLSv1.3	1879	Client Hello (SNI=accounts.google.com)

Can you find any packets that are neither sent by nor destined specifically for your computer?

This can be done through the same method used above to find only my IP address as the source address but with some modifications. The following filter is to be used, **ip.addr != 192.168.100.37**. This filter displays all packets that do not involve my computer at all.

ip.addr != 192.168.100.37						
No.	Time	Source	Destination	Protocol	Length	Info
7675	49.155537	192.168.100.3	239.255.255.250	IGMPv2	64	Membership Report group 239.255.255.250
7677	51.098583	192.168.100.68	224.0.0.251	IGMPv2	50	Membership Report group 224.0.0.251
7679	53.966001	192.168.100.13	224.0.0.251	IGMPv2	50	Membership Report group 224.0.0.251
8604	58.477459	192.168.100.3	224.0.0.251	MDNS	402	Standard query response 0x0000 PTR HP4411-28bba1e31ec6e83ab982f68fee9e
9099	59.500624	192.168.100.3	224.0.0.251	MDNS	402	Standard query response 0x0000 PTR HP4411-28bba1e31ec6e83ab982f68fee9e
10598	61.032663	192.168.100.1	239.255.255.250	SSDP	174	M-SEARCH * HTTP/1.1
10700	61.292495	192.168.100.1	239.255.255.250	SSDP	174	M-SEARCH * HTTP/1.1
10787	61.445269	192.168.100.3	224.0.0.251	MDNS	402	Standard query response 0x0000 PTR HP4411-28bba1e31ec6e83ab982f68fee9e
10828	61.552373	192.168.100.1	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
10856	61.812712	192.168.100.1	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
12033	68.210908	192.168.100.68	224.0.0.251	MDNS	184	Standard query 0x0004 PTR _googlecast._tcp.local, "QM" question PTR _3
12034	68.210908	192.168.100.3	224.0.0.251	MDNS	402	Standard query response 0x0000 PTR HP4411-28bba1e31ec6e83ab982f68fee9e
12035	68.210908	192.168.100.3	224.0.0.251	MDNS	417	Standard query response 0x0000 PTR HP4411-28bba1e31ec6e83ab982f68fee9e
12036	68.214289	192.168.100.3	224.0.0.251	MDNS	417	Standard query response 0x0000 PTR HP4411-28bba1e31ec6e83ab982f68fee9e
12041	69.121489	192.168.100.13	224.0.0.2	IGMPv2	50	Leave Group 224.0.0.251
12220	77.519033	192.168.100.68	224.0.0.2	IGMPv2	50	Leave Group 224.0.0.251

Why do you see packets not sent or received by your computer?

Most of the following protocols use broadcast/multicast communication which means packets are sent to multiple devices rather than a single destination. Examples are (Wireshark Wiki, 2026):

IGMP – Used to manage multicast group membership such as devices trying to join/leave multicast groups.

mDNS – it is used to discover local network devices and also for them to discover each other the local network.

SSDP – used for device/service discovery by sending multicast messages across the network.

ARP – uses broadcasting to map IP addresses to the corresponding MAC addresses, so all device on the network can see these requests.

4. Classification of Network Traffic.

Pavalaydon

Protocol	%Packets	%Bytes
IPv4 (overall)	95.7	4.1
TCP	86.1	3.8
UDP	9.5	0.2
of which DNS	7.1	1.5
IPv6 (overall)	0.4	0
UDP	0.2	0
DHCPv6	0.2	0

Ghaboos

Protocol	%Packets	%Bytes
IPv4 (overall)	97.7	3.8
TCP	56.5	2.3
TLS	19.9	65.7
UDP	40.8	0.6
of which SSDP	2.5	1.4
of which QUIC IETF	12.2	14.5
of which XML	1.3	1.6
of which DNS	3.5	0.8
of which data	20.7	7.7
IPv6 (overall)	1.9	0.1
UDP	1.5	0
of which XML	1.0	1.3

Immambocus

Protocol	%Packets	%Bytes
IPv4 (overall)	99.5	2.3
TCP	22.6	0.5
UDP	76.8	0.7
of which QUIC IETF	74.7	75.8

Note that negligible traffic has been omitted from the classification so as to meet the requirements of this assignment.

5. Based on the Classification, how can Network Performance be Optimised?

The thing about standards and protocols is, they're like crack, some people can't seem to have enough of them! A standard for this—what's that? Another protocol for this same application? Didn't like it? Don't you worry, "you can just sit back and wait for next year's model" (Tanenbaum, 2003). The multitude of standards and protocols can be attributed to: either the current standard not being up to someone's (literal) standards; the implementation of the new and improved protocol dragging its feet due to the old ways having a deathly grip on the existing infrastructure, or, because, as the Spanish say, of *la política de la manta corta*; that is, protocols serving the same global purpose are used in conjunction as they (in most cases) offer features that are *sacrilegiously* impossible for the others to, like expecting TCP to be speedy, UDP not to lose packets, or like expecting banks to stop using COBOL and dropping in the numerous better suited programming languages out there.

Transport Protocols (TCP, UDP, QUIC IETF)

To quote Tanenbaum (2003) on this matter, "*the debate between connectionless and connection-oriented service has been going for decades and is rooted in deep philosophical differences...*" In the beginning, the specifications made for the building of network services—which are unfortunately outside of the scope of this assignment—gave rise to two conflicting *belligerents*. On one side, we have those [UDP] that argued that no matter the subnet, they are indubitably unreliable and as such error control should be done at the hosts. On the other side, we have those [TCP] that argued that reliable and infallible networks are nevertheless possible, as seen by the 100 years of the successful telephone system behind them. Lastly, in the past years, another side has joined this unending tug o' war, namely QUIC IETF, "designed to combine UDP's speed with added reliability, security and efficiency" (Northcutt, 2025); it is built on UDP and reduces latency and connection overhead while improving security.

Optimising the network performance depends on which side of the fence you fall on; if speed is of the essence, and an unreliable connection does not bother you, upping the percentage of UDP traffic is the way to go; but if you like your network smart and proactive, maybe QUIC IETF can become the new TCP.

IPv4 vs IPv6

It seems like it's been forever that we've been told that IPv6 was coming to replace all our IPv4 addresses, but the thing is, changes take time, particularly in a field that is as averse to change as computer science, where we find ourselves at the whims of the fragile environments built of legacy systems held up with scotch tape and the hope and tears of our forefathers. The more you learn about the field, the more it becomes clear that none of *this* was planned, and that we have spent more time extinguishing fires than anything else. On the subject of IPv6, yes, it is common knowledge that it fixes the address drought that has come with the Internet becoming as massive as it is, but it also simplifies the packet header (making for faster processing by routers), offers better support for options (again speeding up packet processing) and is leaps and bounds better in terms of security (integrated authentication and privacy features). Hopefully, it is only a matter of time before IPv6 becomes the convention.

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